



NovoRetin™

The natural alternative to retinol



Based on mastic, which is an aromatic resin from the bark of a tree that grows on the Greek island of Chios, NovoRetin™ is able to increase the amount of naturally occurring retinoic acid in the skin, which leads to retinol-like effects.

The issue

Retinol is well-known for its strong anti-aging effects and the benefits it offers for acne-prone skin. Retinoic acid is the major active metabolite of naturally occurring retinol in skin. It is important for many biological processes, including cell growth and cellular differentiation, and thus the maintenance of the skin. Retinoic acid is able to rapidly induce its own degradation by upregulating the expression of CYP26A1 enzyme. Counteracting this degradation by topical application of retinoids can lead to irritations and stability in cosmetic formulations is poor.

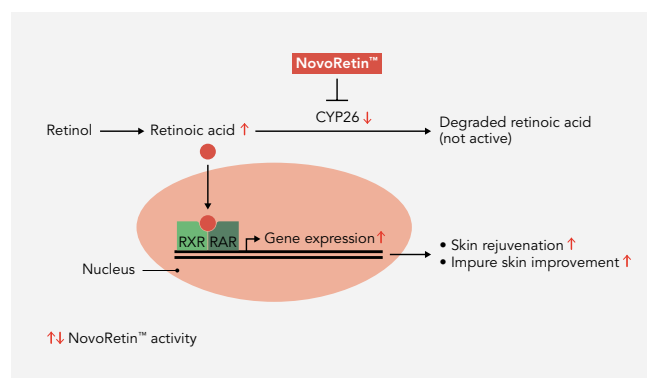
The solution

By inhibiting CYP26 enzymes, it is possible to increase the endogenous levels of retinoic acid. NovoRetin™ is a natural CYP26A1 inhibitor that can prevent the degradation of retinoic acid in the skin. Therefore, by using NovoRetin™, naturally occurring retinoic acid in the skin can be retained, which leads to retinol-like effects in the skin without the need to apply retinoids topically on the skin.

The benefits

NovoRetin™:

- boosts naturally occurring retinoic acid in the skin
- improves skin elasticity and density
- effects immediate lifting
- visibly refines pores and blemishes
- provides a stable and safe retinol activity
- demonstrates clinical skin brightening effects.



The plant

The mastic tree or lentisk (*Pistacia lentiscus*) is an evergreen shrub that grows on arid, rocky and poor soil. Lentisks are typical maquis-type vegetation found in Mediterranean countries. However, only the mastic trees of southern Chios give the typical resin when the bark is scratched. Mastic has been consistently used in skin care formulations for hundreds of years but its application has been limited due to its stickiness and insolubility in water. This is no longer an issue as Mibelle Biochemistry has developed a unique delivery system to make mastic bioavailable for the skin – NovoRetin™.

INCI (EU/PCPC) Declaration

Pistacia Lentiscus Gum / Pistacia Lentiscus (Mastic) Gum (and) Lecithin (and) Pentylene Glycol (and) Glyceryl Caprylate / Caprate (and) Caprylic / Capric Triglyceride (and) Aqua / Water (COSMOS approved*, NATRUE approved).

* Raw material approved by ECOCERT GREENLIFE in accordance with the COSMOS Standard

Study results

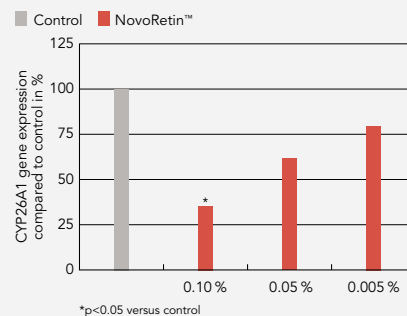
Inhibition of CYP26A1 Gene Expression and Activity

S-872



Cell line: Human keratinocytes (HaCaT)
Test compound/ Treatment: Treatment of cell cultures with test substance in the presence of NovoRetin™ for 9 hours (activity) / 4 hours (gene expression)
Parameters: Activity of CYP26A1 (radiolabelling), gene expression of CYP26A1

- Concentration-dependent reduction of CYP26A1 gene expression
- Concentration-dependent reduction in CYP26A1 activity
- Inhibitory effect on CYP26A1



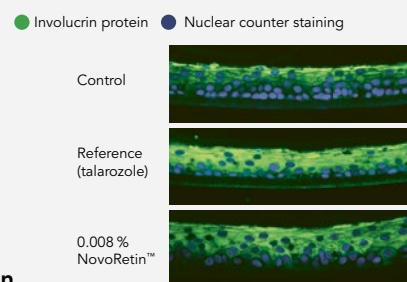
Retinoic Acid Boosting Effect

S-937



Skin model: Reconstructed human epidermis (RHE)
Test compound/ Treatment: Treatment of cell cultures with 0.008 % NovoRetin™ or the reference compound talarozole (CYP26 inhibitor)
Parameters: Involucrin (IVL) gene expression, IVL protein expression (IVL is a marker for retinoic acid activity)

- Significant upregulation of IVL gene expression by +720 %
- Strong upregulation of IVL protein abundance
- Topical application of NovoRetin™ can enhance retinoic acid levels in the skin



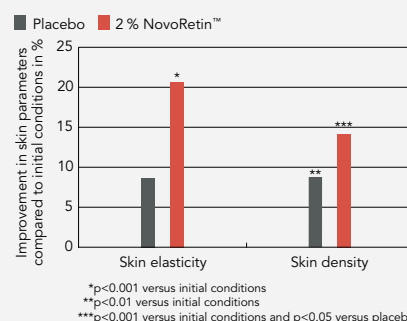
Increased Skin Density & Skin Elasticity

S-1043



Volunteers: 18 (f, 42y–70y) with wrinkles on the crow's feet region and signs of photoaging
Test compound: Cream + 2 % NovoRetin™, placebo
Treatment: Application twice daily on the face and forearms for 28 days
Parameters: Elasticity in the face (Cutometer), epidermis-dermis density on the forearm (Dermalab® Ultrasound)

- Significant improvement of skin elasticity and skin density after 28 days of application



In Addition

S-350/S-778/S-988/S-1122



Improvement in Pore Appearance, Sebum Production and Blemishes: NovoRetin™ significantly decreased pore volume by more than 40 % after 2 weeks. Furthermore, it was shown that NovoRetin™ exhibited a purifying and matifying activity and significantly decreased sebum production, while at the same time increasing skin hydration.



Immediate Anti-Wrinkle Effect: NovoRetin™ significantly improves wrinkle depth, wrinkle length and skin roughness 30 min after a single application.



Skin Brightening Effect: NovoRetin™ significantly and visibly increases skin brightness and even skin tone after 28 days compared to placebo.

Recommended use level: 2–3 %

Characteristics: Water-soluble, without preservatives

Applications: Advanced age-defying formulas, immediate lifting products, treatments for acne-prone skin, pore minimizing formulas

Moreover: Very special and endemic source from the Mediterranean region, 100 % natural origin

Benefits NovoRetin™

- Boosts naturally occurring retinoic acid in the skin
- Improves skin elasticity and density
- Effects immediate lifting
- Visibly refines pores and blemishes
- Provides a stable and safe retinol activity
- Demonstrates clinical skin brightening effects